PCM-HUB

User manual

For linguists

Parametric Comparison Method · Università di Modena e Reggio Emilia

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Contents

1. What is PCM-Hub 5
   - The three roles 5
2. Logging in 6
   - First access 6
   - Accepting the Terms of Use and Privacy Notice 6
   - Changing the password 6
   - Forgot the password? 7
   - Session expired 7
   - Failed-login limit 7
3. Getting around: sidebar, topbar, breadcrumb 8
   - Sidebar (left) 8
   - Topbar (top) 9
   - Footer 9
4. The Dashboard 10
   - User dashboard 10
   - Admin dashboard 10
5. Shared pages 12
   - Glossary (`/glossary`) 12
   - Instructions (`/instructions`) 13
   - How to cite (`/how-to-cite`) 14
   - My Account (`/me`) 15
   - Manual (`/manual`) 15
6. The Languages page 16
   - Filters (they stay at the top) 16
   - The table 18
   - Notes on the map 19
7. Compiling a language 20
   - Language header 20
   - Status banner (below the header) 20
   - Parameter wizard (the squares) 22
   - A parameter's block 23
   - Questions 24
   - Example clipboard 25
   - Saving a parameter block 25
   - Switching parameter without saving 25
   - Submit / Reopen / workflow 26
   - What an admin sees in addition 26

16. Common errors and what to do 27
"Form locked by the current language status" 27
"Question X needs at least 2 valid examples when answering YES or UNSURE" 27
"This block has been modified by another session..." 27
"Wrong email or password" 27
"Too Many Requests" at login 27
"Invalid email format" 27
"You cannot delete your own account" 27
"You cannot delete the last remaining administrator" 27
The "Submit for approval" button is missing 27
The site logged me out unexpectedly 27
I changed the taxonomy but languages still show the old names 28
17. Quick interface glossary 29

Practical guide to using the **PCM-Hub** website (`hub.parametricomparison.unimore.it`). This manual is for daily users: linguists who compile their assigned languages, and administrators who manage parameters, questions, accounts and cross-language analyses.

> **You'll find what you need faster** if you know your role:
>
> - **Linguist (User)** → read sections 1-7 and jump straight to §7 (Compiling a language).
> - **Administrator (Admin)** → read everything: admins see all User functionality plus a dozen extra tools (sections 8-15).

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1. What is PCM-Hub

The **Parametric Comparison Method** (PCM, Longobardi & Guardiano 2009) compares different languages using a set of **universal syntactic parameters**. Every language is "compiled" by a linguist who, for each parameter, answers a set of questions (mostly yes/no) and provides linguistic examples to support the answers.

From the data collected, the website automatically produces:

- consolidated parameter values for each language (`+`, `-`, `0`, `?`);
- comparative matrices across languages;
- Hamming, Jaccard and geographic distances;
- dendrograms and cluster maps;
- PCA analyses, Mantel tests.

This makes it possible to study phylogenetic relationships, divergences and convergences across languages from different families.

The three roles

Role	What they can do
Public (visitor, not logged in)	sees only the home page with the interactive map and the How to cite page
User (assigned linguist)	compiles their assigned languages, accesses glossary, instructions, language list, personal dashboard
Admin	everything a User does, plus: parameters/questions/motivations/taxonomy management, account creation, TableA, queries Q1-Q10, history/backup, migration, recompute

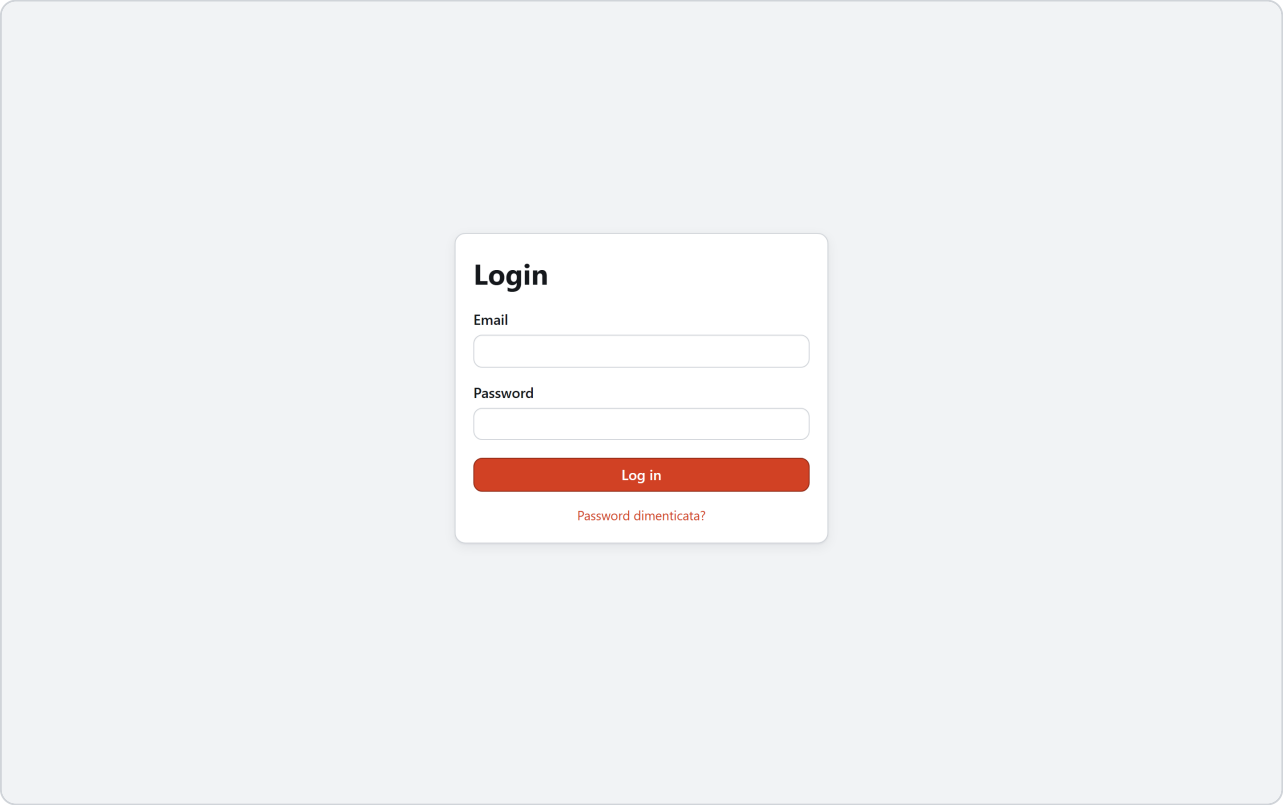
2. Logging in

First access

The administrator who created your account will give you:

- the **email** of your account;
- a **temporary password** (at least 8 characters).

Go to the site URL → click **Login** from the home page (or go straight to `/login`) → enter the credentials → **Log in**.



On your first login **we recommend changing your password**: top-right click **MyAccount** → **Change password**.

Accepting the Terms of Use and Privacy Notice

The first time you log in — and again whenever a new version of the legal documents is published — a modal pops up asking you to **review and accept the Terms of Use and the Privacy Notice** (with a separate checkbox for the "vexatious clauses", where the law requires it). Until you tick the box(es) and click **Accept**, the only alternative is **Logout**: acceptance is mandatory to use the site. The documents are linked inside the modal so you can read them before accepting.

Changing the password

From the **My Account** page you have two blocks:

- **Profile**: name, surname, email (if you change the email it becomes the new login credential).
- **Change password**: requires the current password + the new one twice. Minimum 8 characters.

Forgot the password?

Email reset is now active. On the login page click **Forgot password?** → enter your account email → you'll receive an email with a reset link (valid for 30 minutes). Open it, set a new password, and log in. For privacy, if the address isn't registered the site replies the same way ("if the address exists you'll receive an email"), so it never reveals which addresses exist.

If the email doesn't arrive (check your spam folder) or the link has expired, request a new one or contact an administrator.

Session expired

The session stays valid for a long time, so in day-to-day use you won't be logged out. If it does expire, protected pages automatically send you back to the login. Nothing is broken — just log in again.

Failed-login limit

For security: **after 5 failed login attempts from the same connection within 1 minute**, the system replies with a "Too Many Requests" error for a few seconds. Wait a moment and retry.

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3. Getting around: sidebar, topbar, breadcrumb

Once logged in, the interface has 3 fixed elements:

Sidebar (left)

All the main pages. The visible items depend on your role:

Item	User	Admin
Dashboard	✓	✓
Citation Guidelines / How to cite	✓	✓
Languages	✓	✓
Parameters	—	✓
Parameters Graph	—	✓
Questions	—	✓
Motivations	—	✓
Taxonomy	—	✓
Table A	—	✓
Filters	—	✓
Accounts	—	✓
History & Backups	—	✓
Legal Documents	—	✓
Migration Import	—	super-admin only (red = dangerous)
Backup Restore	—	super-admin only
Instructions	✓	✓
Manual	✓	✓
Glossary	✓	✓

Super-admin: two destructive tools — **Migration Import** and **Backup Restore** — are reserved for administrators flagged as *super-admin*. A normal admin doesn't see these two items at all. **Import from Excel** is super-admin only too: it isn't in the sidebar, it opens from the link at the bottom of the Backup Restore page.

The **button at the top of the sidebar** collapses/expands it (useful to gain horizontal space, especially in TableA). The preference is remembered.

Topbar (top)

- **Breadcrumb** on the left: shows the page path (e.g. `Dashboard / Languages / Italian / Data`). Items in **blue** are clickable; items in **grey** are informational only (e.g. dynamic IDs).
- **MyAccount** on the right: shortcut to the profile page.
- **Theme switch (Light/Dark)**: switches the visual theme. The preference is remembered too.
- **Logout**: ends the session and brings you back to the public home.

Footer

At the bottom of every page:

- PCM-Hub citation (CC BY 4.0);
- links to Privacy Policy, Disclaimer, lab contact email (`pcm_lab@unimore.it`);
- link to the institutional PCM Unimore project website;
- **Role badge** (bottom right): tells you whether you are “Admin Access”, “User Access” or “Public View”.

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4. The Dashboard

It's the first page you see after login.

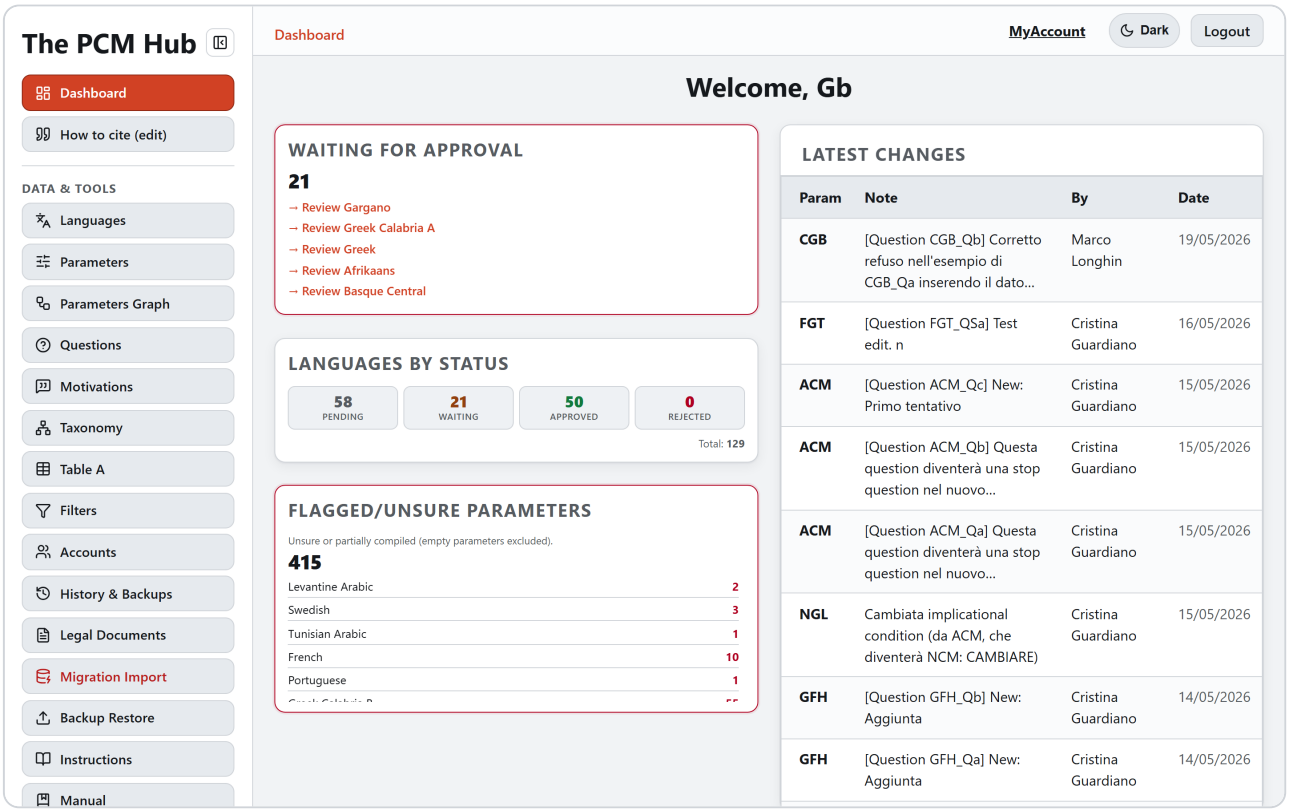
User dashboard

Shows **the languages assigned to you**. For each language:

- ID and full name (e.g. `ITA – Italian`);
- **status** (`Pending`, `Waiting`, `Approved`, `Rejected`);
- **progress bar** (number of answers given / total questions, percentage);
- **Fill in** button (or **View rejection and reopen** if the language was rejected) leading to the compilation page.

If the language is `Rejected`, below you'll also see the **admin's note** explaining why it was rejected.

Admin dashboard



3+1 card layout:

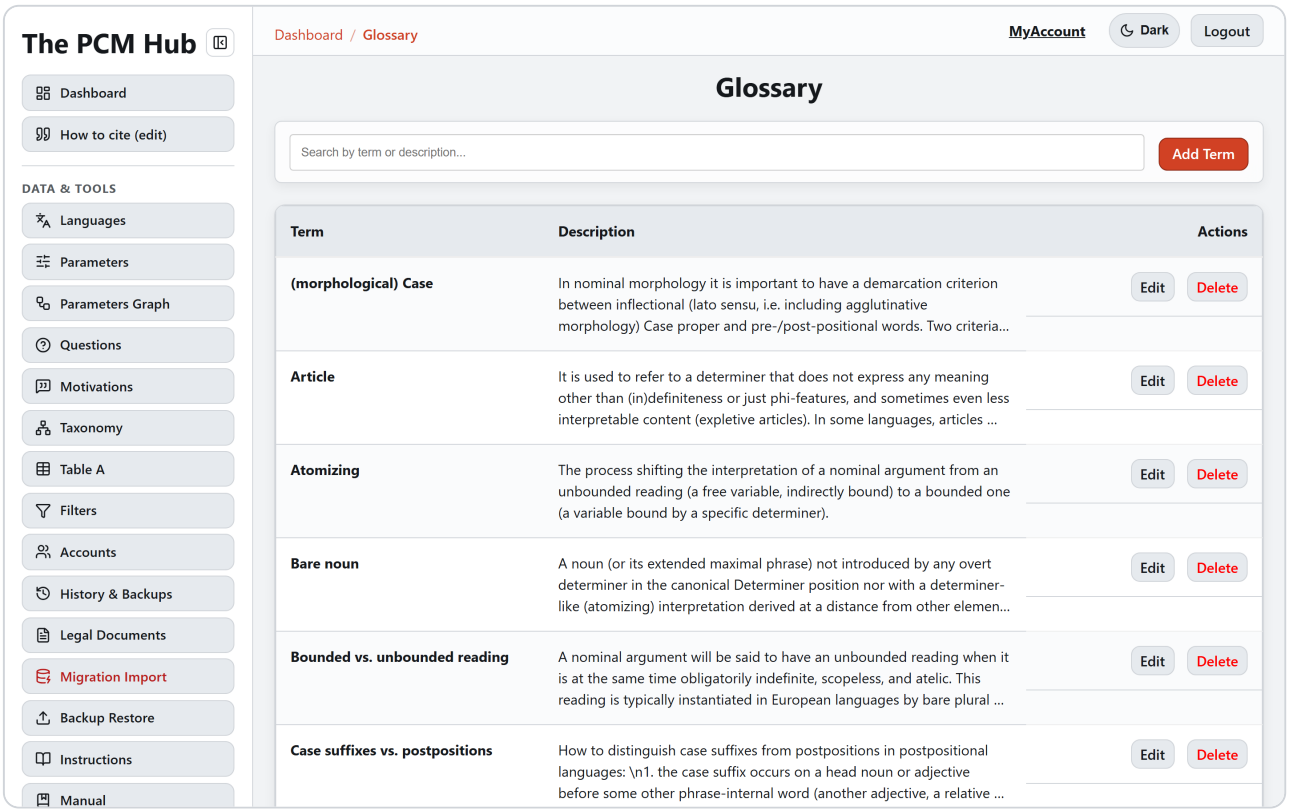
- **Waiting for Approval**: count of languages awaiting review + quick links to open them.
- **Languages by Status**: clickable counters `Pending / Waiting / Approved / Rejected`. Clicking a counter expands a panel listing the languages in that status.
- **Incomplete Parameters**: count of "red" parameters (only partially compiled) grouped by language, with totals.
- **Latest Changes** (right): table with the most recent parameter changes (who, when, what, note).

From here an admin can quickly open languages that need attention.

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5. Shared pages

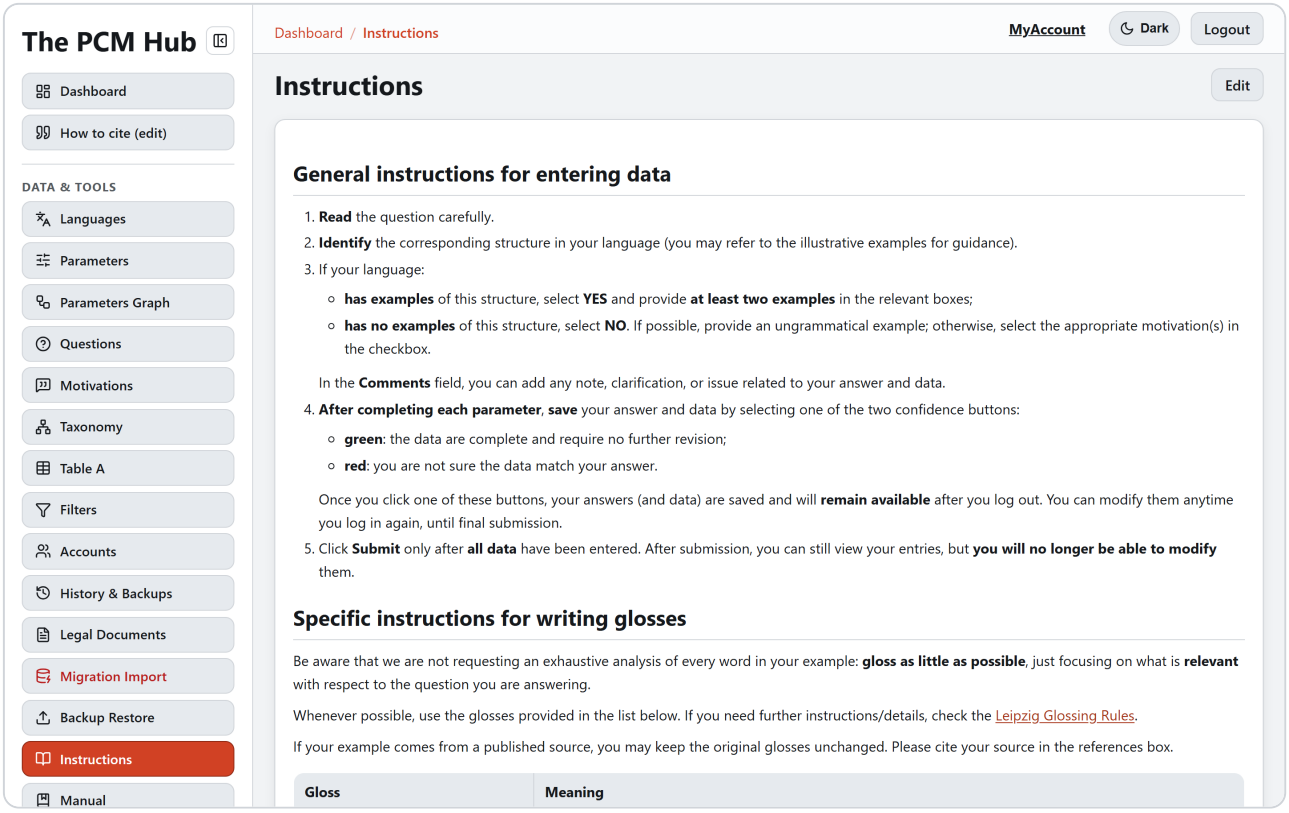
Glossary (`/glossary`)



Alphabetical list of technical terms used in the project, with descriptions.

- All users can **search** (search field at the top).
- Admin only: **Add term**, **Edit** and **Delete** buttons on rows.

Instructions (`/instructions`)

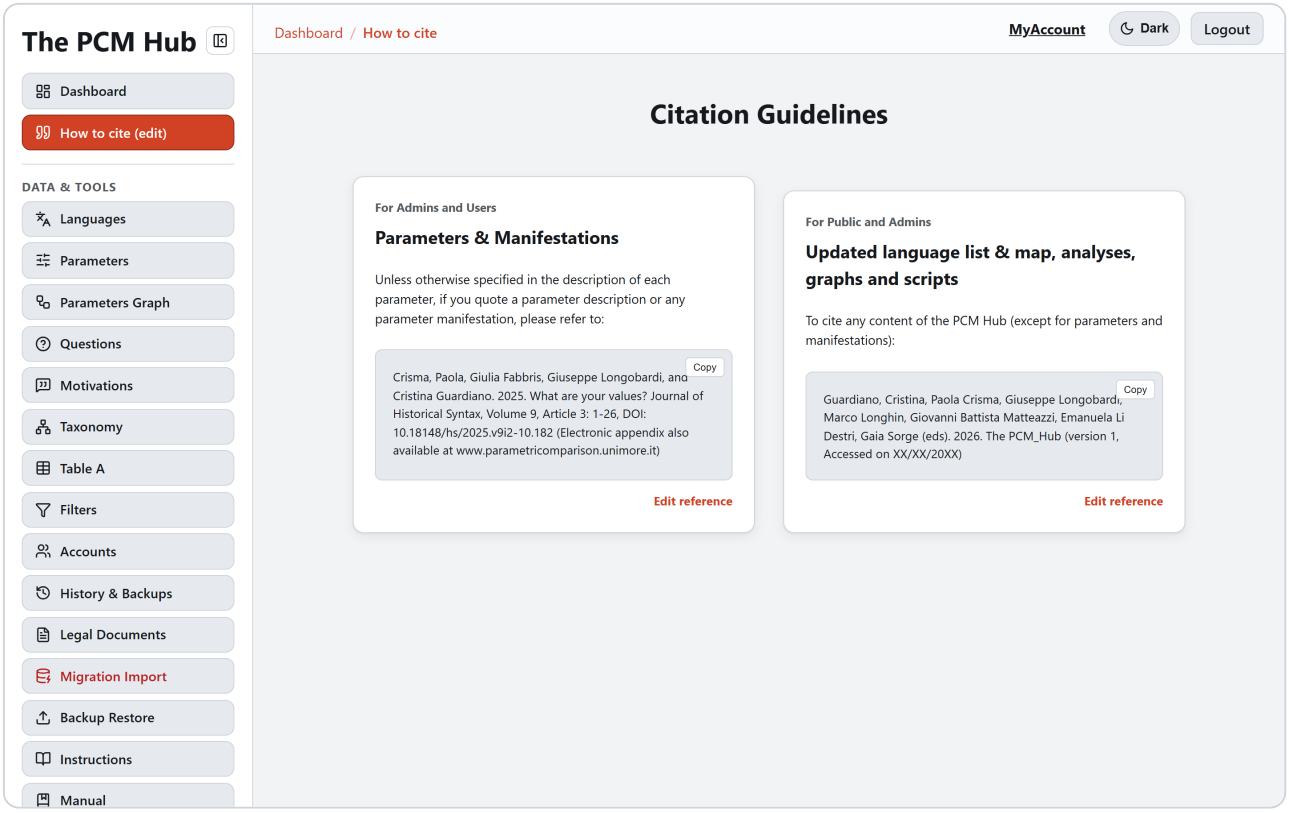


Editable HTML document (powered by TinyMCE) with the compilation guidelines. The default version includes:

- general rules (read → identify → answer YES/NO → save → submit);
- specific instructions for **linguistic glosses** with a symbol table (ACC, NOM, GEN, etc.);
- glossing examples.

Admin only: **Edit** button at top right → a visual editor (you see the result as you type) with formatting, tables, and raw HTML if needed. **Save** persists the document for everyone.

How to cite (`/how-to-cite`)



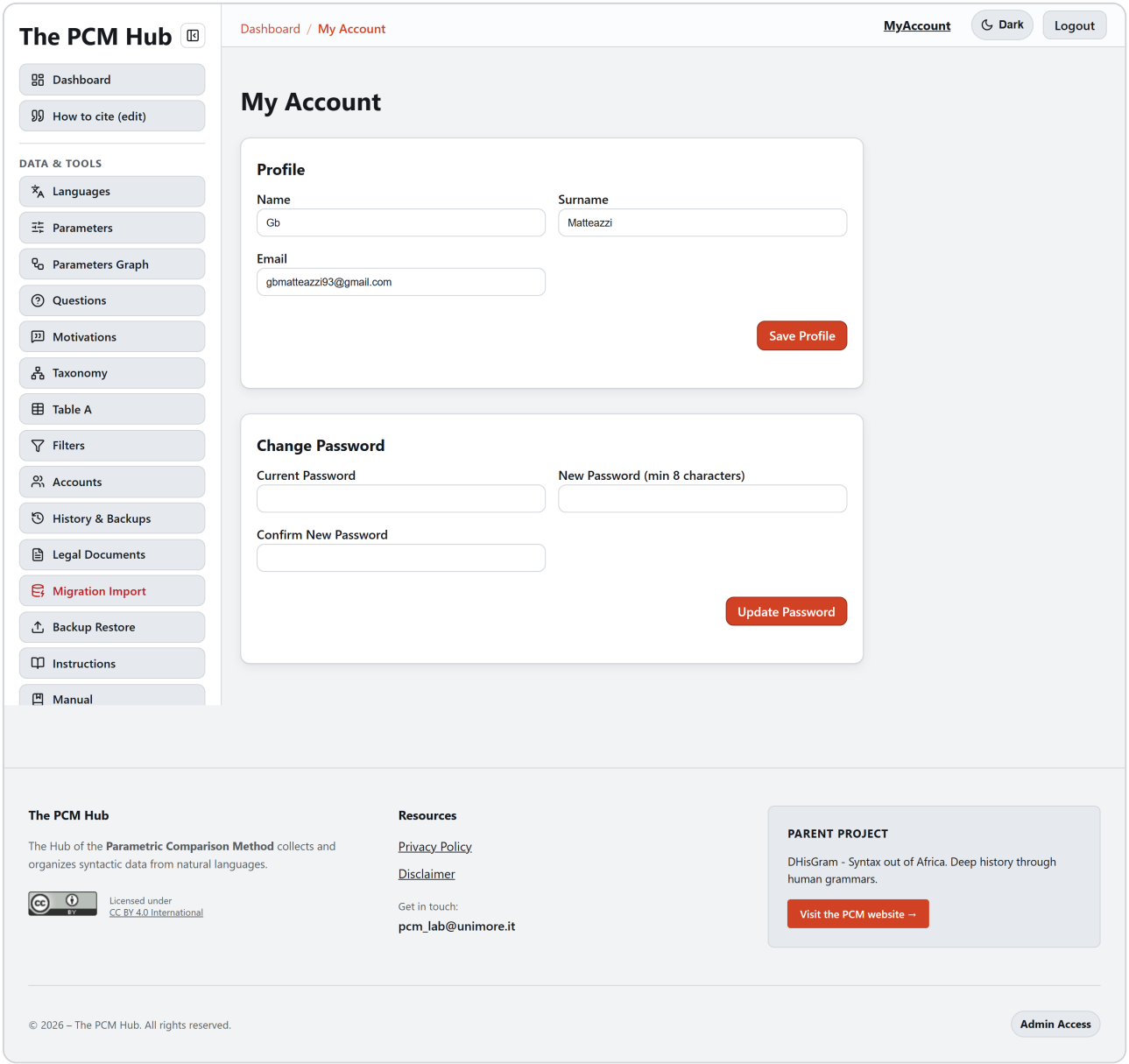
Two cards:

- **Parameters & Manifestations**: how to cite the PCM Hub parameters/questions (recommended for User and Admin).
- **Database** (linguistic data): how to cite the linguistic data collected.

Below each card there is a **Copy** button that copies the full citation to the system clipboard.

Admin only: **Edit** button opens the TinyMCE editor for the citation text.

My Account (`/me`)



- **Profile**: name, surname, email. **Save** writes directly to your profile.
- **Change password**: old password + new one twice. Minimum 8 characters. If the two new copies don't match, the system rejects.

⚠ If you change the email, the **new email becomes your login credential**. Memorize it well (or use a password manager).

Manual (`/manual`)

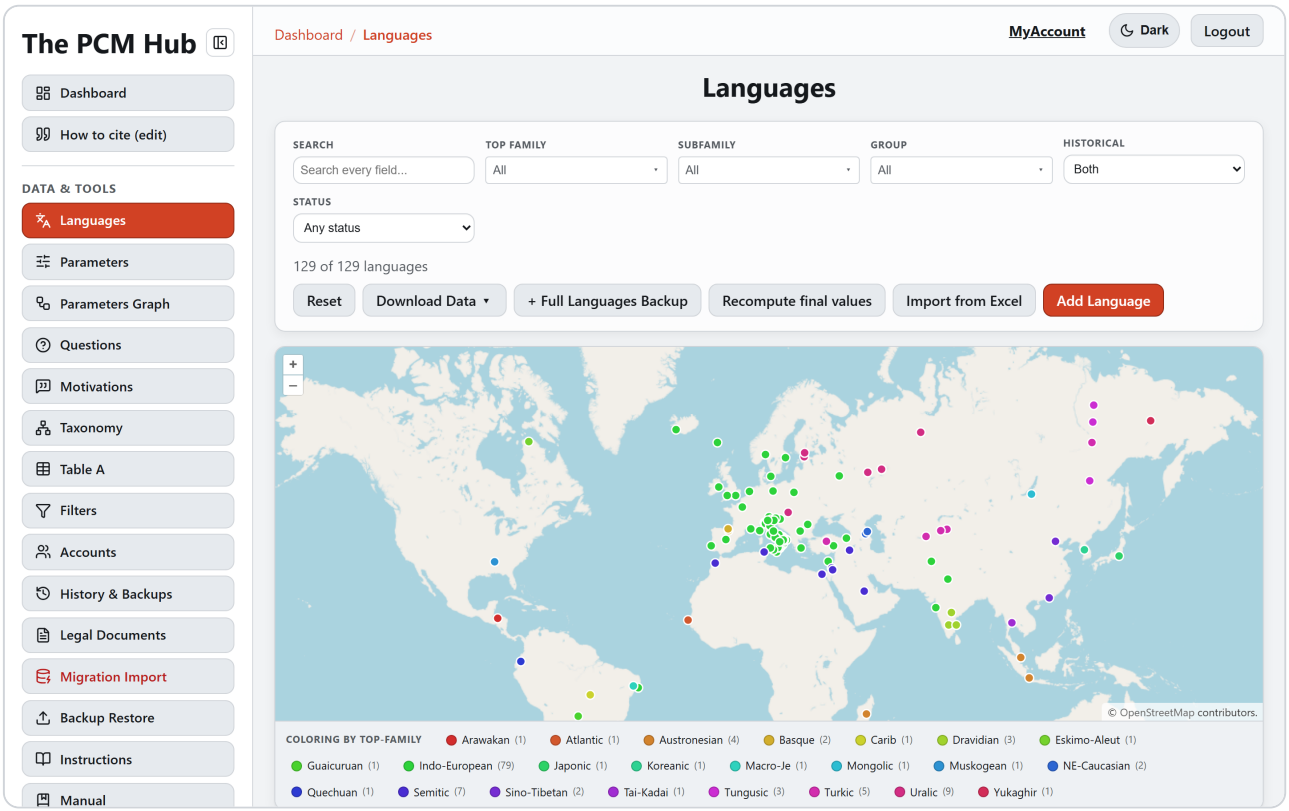
The page where you download the PDF manuals of PCM-Hub. The list adapts to your role: linguists see the **User manual**, administrators see the **Complete manual** (linguists + administrators), each in **English and Italian**. Every entry has a download button — or a "coming soon" label until that PDF has been published.

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6. The Languages page

The list of all the languages in the project. **Above** the table there is an **interactive map** (OpenStreetMap) with a pin for each georeferenced language.

Filters (they stay at the top)



- **Search** global text (ID, name, family, top family, group, status, rejection note).
- **Top Family / Subfamily / Group**: hierarchical multi-select. Choosing a top-family narrows the available subfamilies, and so on.
- **Historical**: all / only historical / only non-historical.
- **Status**: any / Pending / Waiting / Approved / Rejected.

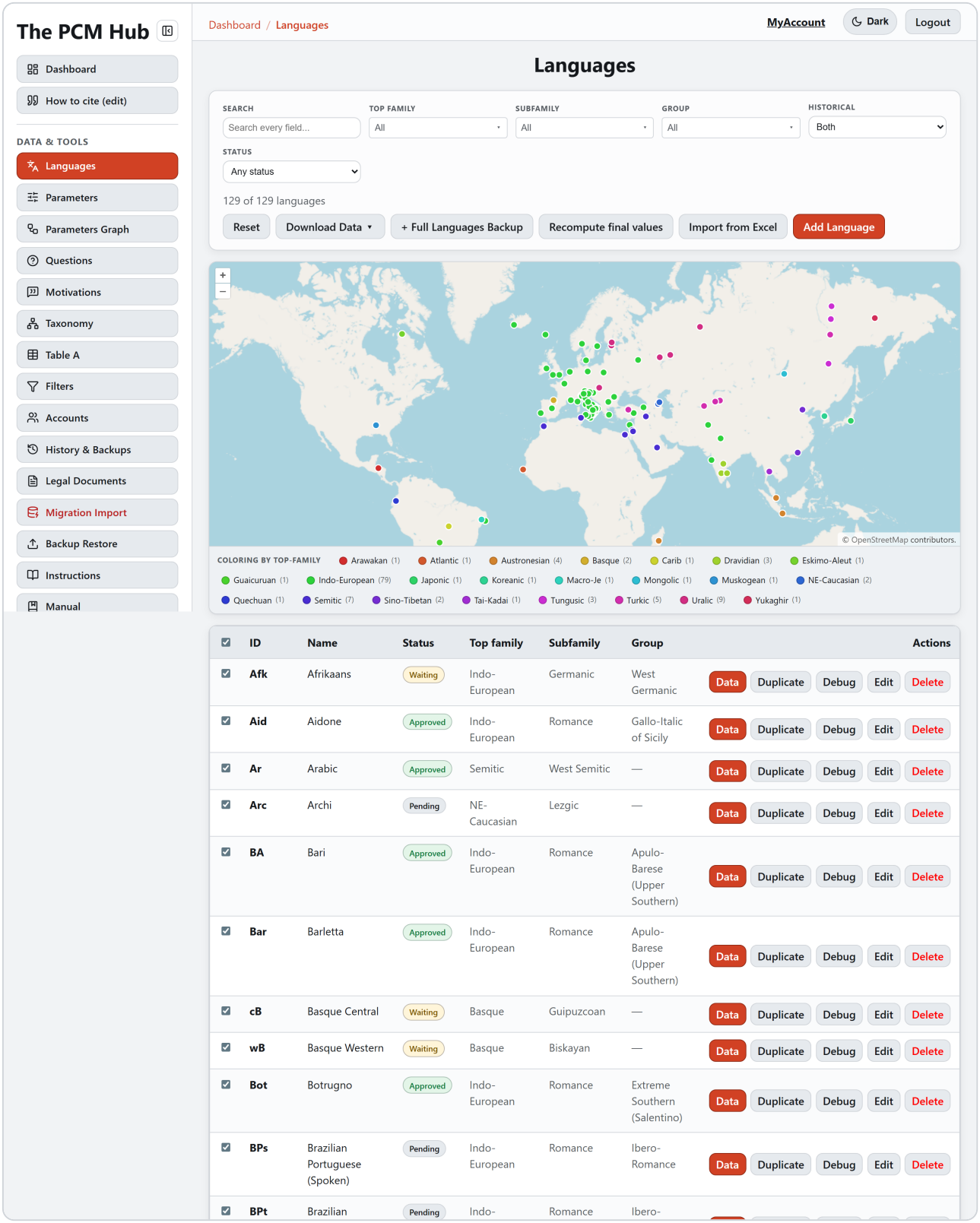
To the right of the filters: counter “X of Y languages”, and the buttons:

- **Reset filters**: clears the search box and all filters.
- **Clear exclusions** (appears if you’ve manually excluded languages).
- **Download** ▾ (admin): opens a menu to export data for the **currently filtered, non-excluded** languages:
  - **Export language metadata (.xlsx)** — ID, name, family, coordinates, etc.
  - **Export backup (.zip)** — full bundle with all answers and examples for the selected languages (this runs in the background: a small notification box at the bottom-right shows progress).
  - **Map (.png)** — the current map as an image.
  - **Geographic distances (.txt)** — GCD km matrix between the selected languages.
- **Maintenance** ▾ (admin): opens a menu with two operations on **all** languages:

- **Recompute final values** — re-runs the implicational DAG. It runs in the background, with a progress bar. Use after important changes to implicational conditions, or after an import.
- **Full Languages Backup** — creates a DB-side snapshot (asks for an optional note). You'll find it later under History & Backups → Full backups.

- **Add Language** (admin): create a new language manually.

The table



- **Left checkbox**: includes / excludes the language from the effective set (map, distances, exports). Ticking the box in the header row excludes/includes all visible languages at once.
- **ID, Name, Status, Top family, Subfamily, Group**.
- **Actions**:

- **Data**: opens the compilation page (regular users and admins).
- **Duplicate** (admin): duplicates the language with all answers and examples (useful as a template). The copy is created with a derived ID that you can change afterwards from **Edit**.
- **Debug** (admin): opens a page showing `init` / `final` values of every parameter for the language, with the option to **Apply implicational condition(s)** (re-runs the DAG on this single language).
- **Edit** (admin): edits the language metadata (name, coordinates, family, etc.). The **ID itself can be renamed** here: when you change it, the old ID is kept as a *historical alias*, so future backup restores and Excel imports that reference the old ID still match this language. The old ID stays free, though: if you give it to another language, from then on it matches that one (the same applies to parameters and questions).
- **Delete** (admin): permanently removes the language. A safety prompt asks you to **type the exact language ID** to confirm. It deletes all of the language's operative data (answers, examples, motivation links, parameter values, its own saved backups and historical ID aliases). It does **not** touch the History audit log (a "delete" event is recorded there), the shared Motivations dictionary, or archived answers from removed questions.

Notes on the map

Markers are colored by **top-family**. The legend below the map lists every top-family with its color: hovering a label dims the others (useful to visually isolate a family).

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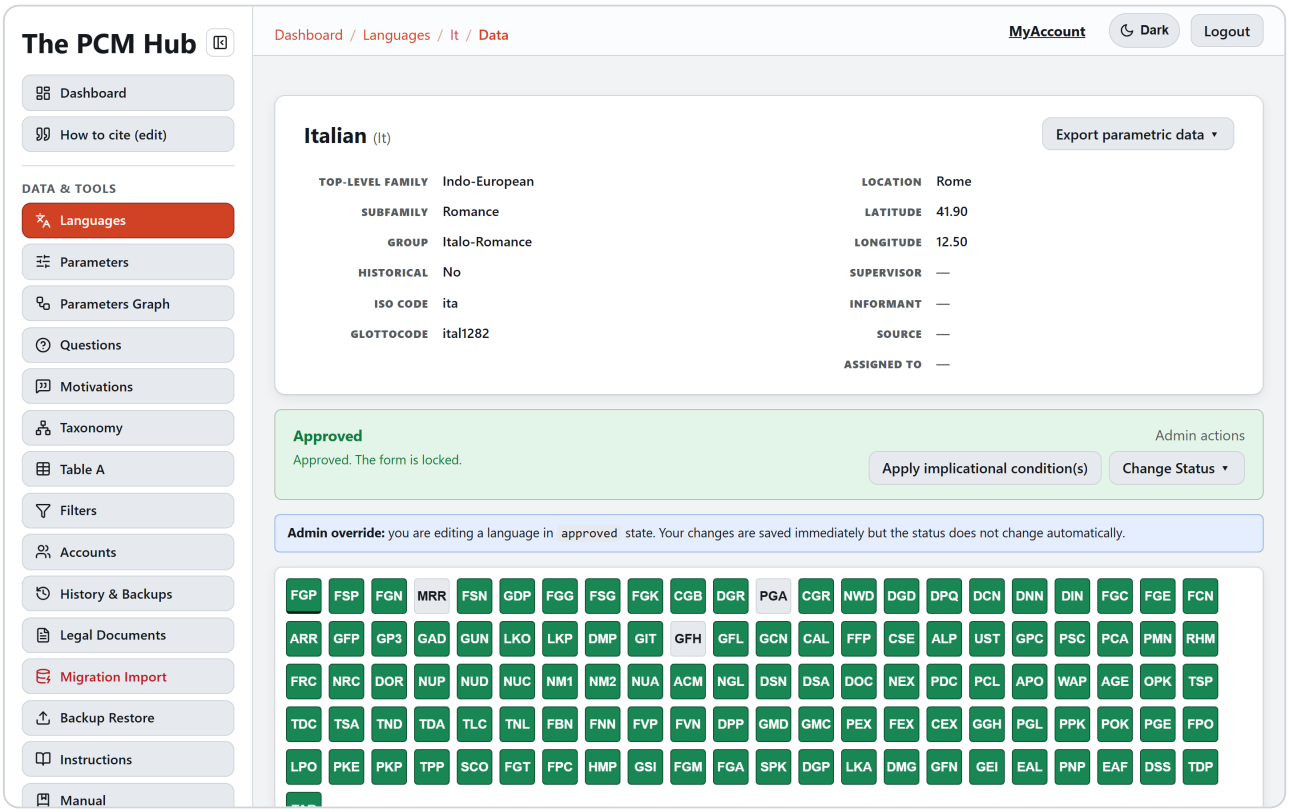
7. Compiling a language

This is the **heart of the site**: where a linguist enters and answers the questions for a language. You get here from:

- Dashboard → click on an assigned language;
- Languages → row → **Data** button.

The URL is `/languages/<id>/data`.

Language header



At the top you see:

- **Name and ID** of the language;
- **Export parametric data (.xlsx)** button (admin) or **Export examples (.xlsx)** (user) to download an Excel with the data you've entered;
- a grid with the language **metadata** (top-family, family, group, ISO/Glottocode, coordinates, supervisor, informant, source, etc.). Only admins also see the assigned user.

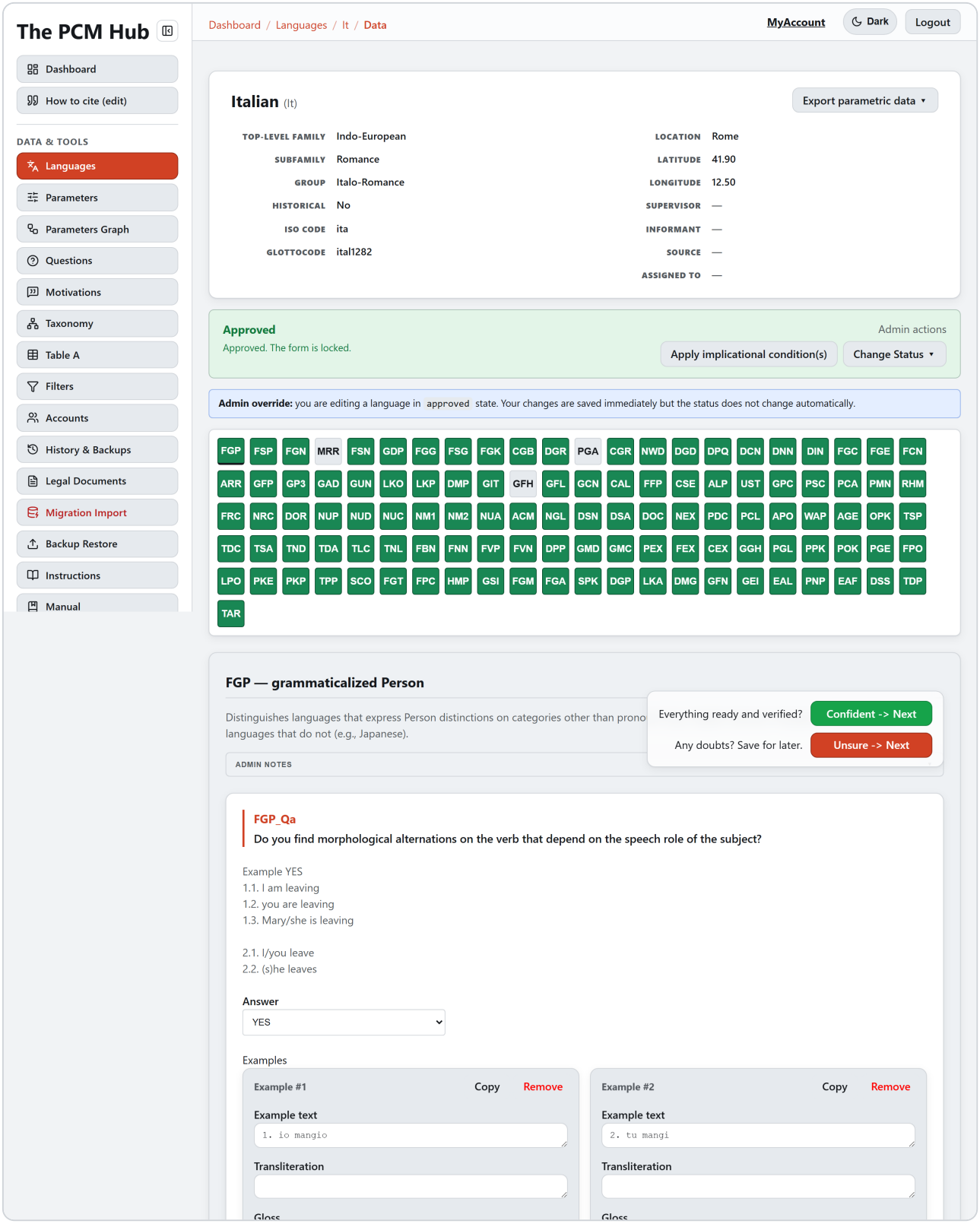
Status banner (below the header)

Shows the current status of the language, a short description and the available workflow buttons. The 4 possible statuses:

Status	Meaning	What you can do (assigned User)
Pending	Being compiled, editable	write/edit freely; when done click **Submit for approval**
Waiting for approval	Submitted, awaiting admin review	the form is **locked**, you have to wait. The admin will receive the language in their Dashboard widget
Approved	Approved, frozen	the form is **locked** read-only. The language joins the official dataset.
Rejected	Rejected, needs revision	read the admin's note, click **Reopen** to put the language back into `Pending` and resume editing

⚠ **Admins instead** can always edit regardless of status (they see a purple "Admin override" banner indicating this). Admins also have a **Change Status** ▾ dropdown that lets them force the language to any status (Approve/Reject/Pending/Waiting), bypassing the normal workflow.

Parameter wizard (the squares)



Below the status banner there is a **row of small squares**, one per active parameter. Clicking a square jumps to **that parameter's question block**.

Squares are colored by the state of the answers:

- **Grey (empty)** — no answers yet.
- **Red (incomplete)** — at least one question of the parameter (stop questions included) has no answer, or is answered `unsure`.
- **Yellow (to re-check)** — every question has an answer, but something needs fixing: a `missing` answer, a `yes` with fewer than 2 examples, an example marked as test, or a question modified by an admin (see ✎ below).
- **Green (complete)** — every question is answered `yes` or `no` and there is nothing to re-check.

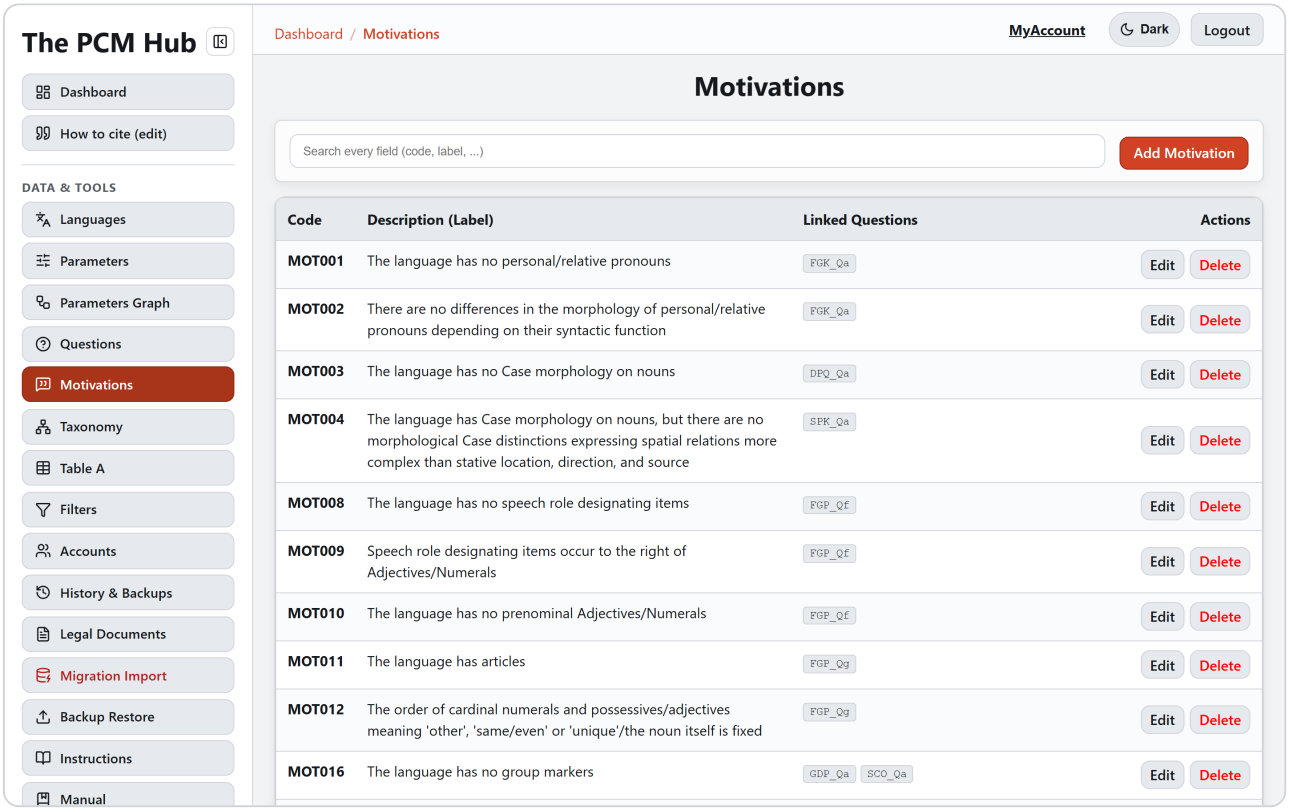
Two small **badges** can appear in the top-right corner of a square, independently of its color:

- **Yellow ✎** — an admin modified a question of this parameter after the language had already answered it. Re-check the block and save it: the badge disappears.
- **Red !** — **answer conflict**: a question and a stop question of the same parameter are both answered `yes`. The two answers contradict each other, so the final value of the parameter usually becomes uncertain (`?`). Fix one of the two answers and save the block: the badge disappears.

When both are present, the red **!** sits just left of the ✎.

Tip: use the colors as a visual to-do list. All green = ready to submit; before submitting, also check that there are no red **!** badges.

A parameter's block



When you click a square you see:

- header with **ID and name** of the parameter, plus a `short_description`;

- (admin only) collapsible **Admin notes** area where the admin can write a free-text note for this (language, parameter) pair. The note is not visible to Users and is not exported.
- **one card per question** in the parameter;
- at the bottom-right a **floating toolbar** (it stays in view as you scroll) with the **Save → Next** save button (green). See below.

Questions

Each question shows:

- a header with the **ID and text** of the question;
- a **More info** button (if there is help text, expandable);
- **Instructions**: general instruction, always visible;
- **Example YES**: an illustrative example, always visible (decorative only, you don't fill it);
- **Instructions (YES)** or **Instructions (NO)**: specific instructions, shown only after you select the corresponding answer;
- **Answer** (dropdown): `— select —`, `YES`, `NO`, `UNSURE`. This is your answer.

Depending on the chosen answer, the card expands differently:

You selected NO

The **Motivations** section appears: a list of checkboxes with predefined reasons why this question is "No" for this language (e.g. "no corresponding structure", "marginal phenomenon"). Tick one or more. If the question has no associated motivations, you'll see an italic message.

You selected YES or UNSURE

The **Examples** block appears: here you must enter **at least 2 linguistic examples** illustrating the answer. For each example:

- **Example text**: the sentence in the language (e.g. *Mario ha letto il libro nuovo che gli ho consigliato*);
- **Transliteration**: transliteration if needed (e.g. for non-Latin alphabets);
- **Gloss**: morpheme-by-morpheme gloss (e.g. *book.M.SG.NOM*);
- **English Translation**: English translation;
- **Reference**: source (e.g. *Hauser, M. 1992. p. 12*).

Useful buttons:

- **+ Add another example**: adds a fresh empty card for a new example.
- **Remove**: deletes an example.
- **Copy**: copies the example to the **internal clipboard** (see below).
- **+ Import example from another answer...**: a search through the examples already written for other questions/languages. Find the example (by text / translation / gloss), click it, and it gets **copied as a new example** in this question. **The imported example is an independent copy**: you can edit it freely without affecting the original.

⚠ **Important rule**: if you answer YES or UNSURE, you **must** provide **at least 2 non-empty examples**. If you try to save with fewer than 2, the system blocks the save with a warning and **highlights** the offending card in red. Same constraint applies to UNSURE.

Comments

At the bottom of each card there is a **Comments** field (free text) for unstructured notes/explanations/doubts.

Example clipboard

Handy utility to avoid retyping the same example in multiple questions of the same language:

1. In any example card, click **Copy** at the top-right → the example goes into the internal clipboard (a red dashed banner appears below your current examples).
2. Move to another question of the same language → you'll see the same banner.
3. Click **Paste here** → the example is added as a **new** example in this question. The clipboard stays full so you can paste into more questions in a row.
4. **Clear** on the banner empties the clipboard.

⚠ The clipboard is **language-specific**. If you switch to a different language, the clipboard is automatically cleared to avoid carrying around an "orphan" example.

Saving a parameter block

At the bottom-right of the current block you see a **floating toolbar** (it stays in view as you scroll) with the **Save → Next** button (green).

The button **saves all data of the current block** (answers + examples + motivations + comments + admin note if you're an admin) and then **automatically advance to the next parameter**.

If the save succeeds: no special feedback, you find yourself on the next parameter.

If the save fails, it can be one of these reasons:

- **400 missing examples**: you put YES or UNSURE on a question with fewer than 2 examples → an alert pops up and the offending card flashes red for a few seconds.
- **409 stale block**: meanwhile someone else (usually an admin) modified the same block. The system warns you and automatically reloads the page so you can review the other person's changes before retrying.
- **Network error**: generic alert, retry later.

Switching parameter without saving

If you've changed something and click another square without saving, the site **asks for confirmation** before discarding the current block's changes. Same if you close the tab or navigate away via breadcrumb: automatic protection.

Submit / Reopen / workflow

Once all the parameters you cared about are compiled, **and** the status is `Pending` or `Rejected`:

- **Submit for approval**: the language goes to `Waiting for approval`. The form **locks** and the admin receives the language in their dashboard.

When the status is `Rejected`:

- **Reopen**: the language goes back to `Pending` (after reading the rejection note). You can resume editing.

Remember: an admin can force the language status at any time and in any direction, via the *Admin override* banner and the **Change Status** dropdown (see *What an admin sees in addition*, below).

What an admin sees in addition

Beyond the purple "Admin override" banner and the **Change Status** ▾ dropdown, an admin opening a language has:

- **Apply implicational condition(s)** (link at the top): opens the language's **Debug** page. For each parameter it shows questions and answers, the **initial value** (from the answers only), the implicational condition with its outcome (TRUE/FALSE) and the **final value**. An orange **!** marks a conflict between a question and a stop question; a **?** with a red **!** marks an uncertain final value, which is also passed on to the parameters that depend on it. The **Legend** button, below "Show inactive questions", opens a compact legend of all the symbols. From the Debug page the admin can also re-run the implicational DAG just for this language.
- **Admin notes** inside each parameter block: free-text note for (language, parameter). It is not exported and not visible to Users.

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16. Common errors and what to do

“Form locked by the current language status”

The language status is `Waiting for approval` or `Approved`, and you are not an admin. Wait for the decision, or (if you are the assigned user and the language is `Rejected`) click **Reopen**.

“Question X needs at least 2 valid examples when answering YES or UNSURE”

You picked YES or UNSURE on a question with fewer than 2 non-empty examples. Add/complete the examples and retry.

“This block has been modified by another session…”

Another user (usually an admin) modified the same block while you were working on it. The page automatically reloads: review the other person’s changes before retrying your save.

“Wrong email or password”

Wrong credentials. If you don’t remember the password, use **Forgot password?** on the login page to receive a reset link by email.

“Too Many Requests” at login

You typed the wrong password 5 times within a minute. Wait 1 minute and retry.

“Invalid email format”

You’re creating an account or changing email with an invalid value (e.g. missing `.it`, space, double `@`). Fix it.

“You cannot delete your own account”

Have another admin do it.

“You cannot delete the last remaining administrator”

Create a second admin **before** deleting the existing one.

The “Submit for approval” button is missing

- Language status is already `Waiting for approval` or `Approved`: nothing to do.
- Status is `Rejected`: see **Reopen**, not Submit.
- You’re not the assigned user for this language: only the assigned user can Submit.
- You’re an admin: admins don’t submit (they go directly through approve/reject).

The site logged me out unexpectedly

Your session expired. Just log in again. (Sessions are long-lived, so this happens rarely.)

I changed the taxonomy but languages still show the old names

The `top_family` / `family` / `group` strings on languages are a stored copy taken from the taxonomy. Reopen each affected language and Save (even without changing anything) to propagate the new names. To do this in bulk: use Excel export/import.

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17. Quick interface glossary

Term	Meaning
Wizard / squares	the row of small buttons at the top of the compilation page, one per parameter
Block / parameter block	the set of all questions of a parameter for one language
Save → Next	the save button of the block: saves everything and moves to the next parameter
Stale block (409)	a block that was modified by another session while you were editing
Submit for approval	sending the language to the admin for review
Reopen	bringing a `Rejected` language back to `Pending`
Approved / Rejected / Pending / Waiting	the 4 states of a language
Admin override	the purple banner reminding the admin they're editing a locked language
Force status	the admin dropdown to override a language's status bypassing the workflow
Super-admin	an administrator with extra privileges — the only role that can access Migration Import, Backup Restore and Import from Excel
Implicational DAG	the parameter dependency graph (`+SPK & -DEM`) deciding whether a value should be neutralized
value_orig vs value_eval	`orig` = aggregated from the linguist's answers; `eval` = `orig` post-DAG (can be `0` if neutralized)
Comparable	a parameter is "comparable" between two languages if both have a non-null `value_eval`
Implicational condition	the string like `+SPK & -DEM` that triggers neutralization
Neutralized	a parameter forced to `0` because its implicational condition is not satisfied
Unsure answer	an `unsure` answer: the parameter's square turns red
Answer conflict	a question and a stop question of the same parameter are both answered `yes` (red **!** badge on the square, orange **!** on the Debug page)
Red parameter	in the Admin Dashboard, "red" parameters = unsure or incomplete
Hamming / Jaccard distance	distance metrics between languages based on parameter values
GCD	Geographic Distance, kilometric distance computed from lat/lng
Mantel test	statistical test measuring correlation between distance matrices

Glossary	dictionary of technical terms
Instructions	the page with compilation guidelines
How to cite	the page with the PCM Hub bibliographic citations
ZIP bundle	the compressed archive used for migration / backup-restore
Recompute	the job that re-runs the DAG over all languages

For doubts, suggestions or bug reports: write to `pcm_lab@unimore.it` (link at the bottom of every page on the site).